

INTRA-PRODEX

Autonomous Fire Detection & Suppression System

Group-5

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Abstract

Traditional automated fire suppression systems often rely on a purely reactive, "one-size-fits-all" approach, deploying a single extinguishing agent - typically water or carbon dioxide - regardless of the fire's chemical origin. This uniform response can be highly inefficient and even catastrophically dangerous.

To address this limitation, this project details the design of an autonomous fire detection and suppression system that bridges the gap between hazard identification and tailored extinguishment. By utilizing a multi-sensor network of MQ-2, MQ-135, and IR sensors interfaced with an ESP32, the system analyzes real-time environmental data, such as combustible gas concentrations, smoke profiles, and thermal signatures to accurately classify the specific type of fire.

Once the fire class is identified, the processing units command a motorized pan-tilt turret to physically target the heat source and automatically deploy the exact extinguishing agent chemically suited for that specific hazard.

The subsequent sections of this documentation provide a comprehensive blueprint of this solution, including an in-depth technical breakdown of the hardware integration, the software's decision-making logic, and the mechanical automation. Finally, it presents a comparative analysis demonstrating how this intelligent, classification-based system offers distinct safety enhancements, superior performance metrics, and a significant reduction in collateral damage compared to conventional suppression methods.

Introduction

The Ubiquitous Hazard of Fire Accidents

Fire accidents represent one of the most unpredictable and devastating hazards in both residential and industrial environments. In industrial and laboratory settings, the presence of volatile chemicals, high-voltage electrical equipment, and complex machinery significantly amplifies this risk. A minor spark or a localized short circuit can escalate into an uncontrollable blaze within a matter of minutes.

The traditional approach to mitigating these hazards heavily relies on human intervention or generic, reactive automated systems. While smoke alarms and sprinkler networks are standard safety features, they often fall short in complex emergencies. Manual suppression requires a human to be present, accurately assess the situation, and risk their own safety to extinguish the flames.

Conventional automated systems typically deploy a single extinguishing agent-most often water-across a wide area as soon as heat or smoke is detected. This generic response leads to the core problem already addressed in this project: not all fires are the same, and treating them uniformly can be inefficient, ineffective, or actively dangerous.

The Critical Necessity of Fire Classification

The fundamental chemistry of combustion varies drastically depending on the fuel source. Consequently, applying the wrong suppression method is not merely ineffective-it can trigger catastrophic secondary reactions. The necessity of classifying a fire before attempting to extinguish it is rooted in these severe risks:

- **Electrical Hazards:** Deploying a conductive agent like water or standard foam onto live electrical equipment creates an immediate, fatal electrocution hazard for anyone in the vicinity and can cause further short circuits, sparking new fires.
- **Boil-Over and Spread:** When water is introduced to a flammable liquid fire (such as oil or gasoline), the water instantly vaporizes into steam upon hitting the high-temperature liquid. This rapid expansion violently scatters the burning oil, aggressively spreading the fire over a larger area.
- **Chemical Reactions:** Combustible metals burn at extremely high temperatures. Applying water to a metal fire strips the oxygen from the water molecules, effectively feeding the fire and potentially causing an explosive reaction.

Therefore, identifying the chemical nature of the blaze and its class is of absolute significance for safe and effective suppression.

Standard Classes of Fire and Corresponding Extinguishing Agents

To standardize safety protocols, fires are universally categorized into distinct classes based on their fuel source. Understanding these classes dictates the selection of the extinguishing agent:

	Fuel	Extinguishing Agent
Class A (Ordinary Combustibles)	Wood, Paper, Cloth, Trash and Plastics	Water or monoammonium phosphate (ABC Dry Chemical)
Class B (Flammable Liquids and Gases)	Gasoline, Petroleum Oil, Paint, Propane, Butane	CO ₂ , or Foam, Specialized Dry Chemicals.
Class C (Electrical Fires)	Energized Electrical Equipment, Appliances, Wiring	CO ₂ , or Specialized Dry Chemicals
Class D (Combustible Metals)	Magnesium, Titanium, Potassium, Sodium etc	Special Dry Powder Agents (like Sodium Chloride-based Powders)
Class K/ F (Cooking Oils and Fats)	Animal and Vegetable fats used in commercial cooking.	Wet Chemical Extinguisher containing Potassium Acetate.

News and Statistics:

Headlines of Hazard

The empirical data and recent case studies surrounding fire incidents underscore a sobering reality: despite advancements in modern infrastructure, fire hazards continue to exact an unacceptable toll on human life and economic stability.

These figures validate the urgent necessity for the intelligent, classification-based automated turret proposed in this project—a system explicitly designed to intercept these precise failures before they escalate into the statistics of tomorrow. Some news following news articles highlight the horrors of fire hazards from all over the world:

- **TikToker demonstrates how not to put out a grease fire in a wild video: ‘We are very embarrassed’:** A viral kitchen incident showed a person attempting to extinguish a grease fire with water, which caused a large fireball explosion and second-degree burns.
- **Scary video proves why you should NEVER pour water on a kitchen blaze as firefighters show ‘massive’ flames:** As firefighters approached a burning pot on a stove top, they used a small amount of water, but just milliseconds later the kitchen fire turned into a massive blaze.

- **TRP Game Zone Fire, Rajkot, India (May 2024):** A devastating fire erupted in a temporary children's play zone, claiming the lives of at least 27 people - including a newlywed couple and four children under 12.
- **CO₂ Fire Suppression System Fatalities, China:** Ten workers were killed and 19 injured when a CO₂ fire suppression system was inadvertently discharged on a cargo ship in Shandong province, China. The incident underscores the danger of deploying the wrong suppression agent.

Statistical Analysis of Fire Incidents and Escalation Rates

Before examining the technical implementation of the proposed automated turret, it is crucial to quantify the scale and severity of the problem it aims to solve. The following section compiles recent empirical data and statistical analyses of fire-related accidents, highlighting a critical point of failure in modern safety infrastructure: the catastrophic consequences of delayed responses and improper extinguishing methods. Ultimately, this data serves to underscore the pressing, real-world necessity for the intelligent, classification-based suppression system detailed in this report.

Global Statistics:

- In 2024, 1.39 million fires in the United States alone caused 3,920 deaths and \$19.1 billion in damage. Home fires accounted for 75% of civilian fire deaths and injuries.
- Over 60% of fire-related deaths happen in the home environment, where accidents like misclassifying fire or late response to fire might occur, according to NFPA data.
- Electrical and lighting equipment accounts for 24% of all structural fires, and also leads to the largest number of injuries and property losses - at 9% and 55% respectively.
- Only 38% of UK consumers claim to know which fire extinguisher to use for each class of fire. When tested, 85% could not identify how to properly extinguish a gas fire, and more than 67% did not know how to correctly put out a cooking oil fire.

India Statistics:

- Electrical short circuits remain the leading trigger of fires in India - responsible for approximately 70% of fires - due to overloading and poor wiring.
- Nearly **20%** of all fire accidents in India occur due to **gas cylinder bursts or stove explosions**, while around **21%** happen due to **electrical short circuits**.
- **LPG** is used in about **62%** of Indian households, while **firewood and solid fuels** are used by approximately **33 to 40%** of residential kitchens - making fire-type classification directly critical to the majority of the Indian population.

Existing Solutions

The following section reviews these standard systems, analyzing their operational methodologies and, critically, their inherent limitations. By examining the shortcomings of these established technologies-particularly their inability to dynamically classify fire types and prevent secondary chemical hazards-we can clearly contextualize the necessity and engineering value of the intelligent, dual-action system presented in this report.

1. Ceasefire Mini – Micro Environment Fire Suppression System



- Utilizes a heat-sensitive glass bulb that breaks automatically when the surrounding temperature exceeds a preset limit (e.g., 57 °C).
- The mechanical breakage instantly triggers the release of a clean extinguishing agent that spreads over the fire.
- Detects only temperature rises, lacking gas sensors to identify the specific fire class.
- Relies on a single, non-adaptive suppression agent.

2. ANSUL R-102 Restaurant Fire Suppression System (India)



- Senses high heat using a fusible link that melts when temperatures reach a critical threshold.
- Activates a release mechanism to discharge ANSULEX liquid agent directly onto the targeted appliances.
- Restricted to very specific hazard environments (e.g., commercial kitchen fires).
- Lacks the capability to dynamically analyze gases or classify the fire type prior to deployment.

3. FirePro Condensed Aerosol Generator (FirePro Systems, Italy/Global)



- Employs a solid fire suppression compound (FPC) that undergoes a rapid reaction when activated electrically or by heat.
- Produces a rapidly expanding aerosol cloud composed of potassium-based particles.
- Extinguishes the fire by chemically interrupting the combustion chain reaction without significantly reducing oxygen levels in the room.
- Deploys a fixed, uniform suppression agent (aerosol) regardless of the specific chemical fire involved.

4. Agni Controls (Flame Detection Based Fire Suppression Systems, India)



- Scans industrial environments for the presence of actual fire using infrared or ultraviolet flame detectors.
- Releases a pre-configured agent, such as CO₂ or inert gas, through a fixed nozzle network into the protected area.
- Relies strictly on visual flame detection without analyzing the ambient gas composition.
- The suppression agent is pre-selected and static, offering no adaptive response to different fire classes.

PRODUCT COMPARISON

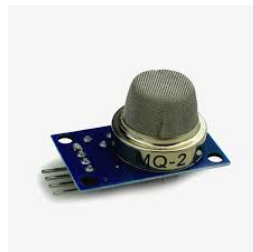
	Fire Classification	Automated Extinguishing
Ceasefire System	✗	✓
ANSUL R-102 System	✗	✓
FirePro Aerosol System	✗	✓
Agni Control System	✗	✓
Our Solution	✓	✓

Proposed Solution

The proposed solution is an Autonomous Fire Detection and Suppression System designed to overcome the limitations of generic, reactive suppression methods. It utilizes a multi-sensor network, including MQ-2, MQ-135, and IR sensors, interfaced with an ESP32 microcontroller to analyze real-time environmental data and accurately classify the specific chemical class of the fire. Once the fire class is identified, the ESP32 serves as the real-time execution unit, commanding a motorized pan-tilt turret to precisely aim at the localized heat source and automatically deploy the extinguishing agent chemically suited for that specific hazard, thus preventing secondary damage and ensuring effective suppression.

Component Description

1. MQ-2 Gas Sensor:



The MQ-2 Gas Sensor is a semiconductor gas sensor primarily used for detecting flammable gases and smoke. It is particularly sensitive to gases such as LPG, propane, methane, hydrogen, and smoke particles produced during combustion. The sensor operates based on the change in resistance of a tin dioxide (SnO_2) sensing layer when exposed to combustible gases.

In this project, the MQ-2 sensor is used to detect the presence of flammable gases and smoke generated during fire events, which helps in identifying gas-related fires.

2. MQ-135 Gas Sensor:



The MQ-135 Gas Sensor is a metal-oxide semiconductor gas sensor primarily used for detecting solid carbon-containing combustible materials. It is particularly sensitive to gases such as carbon monoxide, carbon dioxide, ammonia and benzene. It detects gases by measuring resistance changes in a heated tin-oxide sensing layer.

In this project, the MQ-135 sensor helps detect combustion byproducts produced during solid fuel fires such as wood, allowing differentiation between gas-based and solid-fuel fires.

3. Infrared Flame Sensor:



IR flame sensor detects infrared radiation emitted by flames. The sensor detects radiation from the flames in the 760 nm to 1100 nm wavelength range. IR flame sensors can detect fire, candle flame, gas burner flame, lighter flame, burning paper, burning wood, and burning alcohol. In this system, the infrared sensor confirms the presence of an active flame, improving reliability by distinguishing actual fire from gas presence alone.

4. ESP-32



The ESP32 serves as the primary hardware interface and real-time execution unit of the system. The ESP32 is responsible for translating those classifications into immediate physical action. It directly controls the servo mechanisms of the motorized pan-tilt turret, ensuring precise aiming at the localized heat source, and triggers the mechanical deployment of the specifically chosen extinguishing agent. Additionally, the ESP32 can facilitate rapid, low-latency sensor polling, efficiently bridging the gap between the environmental hardware and the central processing unit.

5. Servo Motor:



A servo motor is a rotary actuator that uses Pulse Width Modulation (PWM) signals and an internal closed-loop circuit for precise angular positioning.

In this project, servo motors drive the automated turret's pan-tilt mechanism. Upon hazard detection, the central processing unit sends PWM signals to the servos, enabling them to dynamically sweep and accurately aim the extinguishing nozzle directly at the identified heat source prior to deploying the agent.

6. LEDs



Visual feedback is managed by a dedicated LED and three RGB modules for real-time status. The primary LED indicates the specific fire classification through distinct flashing patterns upon detection. Meanwhile, the RGB LEDs provide secondary indications: one monitors system readiness, the second confirms target acquisition once the servos lock onto a heat source, and the third signals safety alerts or low agent levels.

7. Battery + Voltage Divider



The system is powered by a 9V battery integrated with a voltage divider circuit to regulate the supply. By utilizing a specific resistor ratio, the 9V input is stepped down to a stable 5V output. This ensures the microcontroller and sensors operate within their safe voltage thresholds for consistent performance.

Product Working

The proposed autonomous fire suppression turret operates through a sophisticated, tightly integrated loop of environmental sensing, data processing, and mechanical actuation. The system architecture is divided into three primary subsystems: the Fusion Sensing, the Dual-Microcontroller Processing Unit (ESP32), and the Electromechanical Turret Assembly. The following sections detail the flow of data and the specific operational mechanics of each subsystem.

The Sensing Phase: Hazard Detection and Data Acquisition

The operational cycle begins with continuous environmental monitoring. The system does not rely on a single parameter; instead, it uses sensor fusion to gather a comprehensive chemical and thermal profile of the area.

- **Thermal Confirmation (IR Sensor):** The Infrared Flame Sensor acts as the primary interrupt or trigger for the system. Tuned to detect wavelengths between 760 nm and 1100 nm, it continuously scans for the specific thermal signature of an active flame. When a flame is

detected, the sensor outputs a digital HIGH signal to the microcontroller, confirming that an actual fire is present, rather than just a gas leak or ambient smoke.

- **Chemical Profiling (MQ-2 and MQ-135):** Simultaneously, the gas sensors analyze the ambient air. The MQ-2 monitors for flammable gases (like LPG and methane) which indicate a potential Class B (flammable liquid/gas) fire. The MQ-135 monitors for CO-2 and specific combustion byproducts typical of burning solid fuels (wood, paper), indicating a Class A fire.

Data Processing and Fire Classification

To handle both the decision-making logic and strict real-time hardware control, the system employs a streamlined, single-microcontroller architecture utilizing the ESP32.

- **Central Intelligence and Data Acquisition:** The ESP32 acts as the central brain of the system. Utilizing its onboard Analog-to-Digital Converters (ADCs), it directly reads the analog signals from the environmental sensors, completely eliminating the need for external ADC modules like the MCP3008. The microcontroller continuously polls these gas values. Once the IR sensor triggers a "Fire Active" state, the ESP32's embedded firmware evaluates the incoming data from the MQ sensors. By comparing the relative concentrations of gases detected by the MQ-2 and MQ-135 against pre-calibrated thresholds, the algorithm classifies the fire. For example, a high MQ-135 reading combined with a low MQ-2 reading classifies the hazard as a Class A solid-fuel fire, while a sharp spike in the MQ-2 classifies it as a Class B chemical fire.
- **Real-Time Hardware Execution:** A major advantage of utilizing the ESP32 for the entire process is its ability to execute code without the overhead of a heavy operating system, avoiding the micro-stutters that can disrupt precise motor control. Immediately after the logic classifies the fire and calculates the required targeting coordinates, the ESP32 seamlessly transitions into its hardware controller role. It directly drives the servos of the pan-tilt turret to aim at the heat source and executes the physical suppression by deploying the correct extinguishing agent, ensuring a rapid, zero-latency response.

Remote Notification and Evidence Capture

Upon fire classification, the ESP32 immediately transmits a serial trigger signal to a connected laptop via USB, identifying the detected fire type (e.g., "**FIRE_GAS**" or "**FIRE_WOOD**"). A Python script running on the laptop continuously listens on the designated COM port for this trigger. Once received, the script activates the system's notification and evidence pipeline:

- The laptop's webcam immediately begins recording, capturing a 5-second video clip of the fire scene, including 2.5 seconds of pre-trigger buffered footage and 2.5 seconds post-trigger.
- The recorded clip, along with the classified fire type and timestamp, is automatically dispatched to the designated user via a Telegram bot, providing real-time remote notification of the hazard.
- This ensures that even in an unattended environment, the responsible personnel are instantly alerted with visual evidence of the fire event, enabling rapid manual intervention if required.

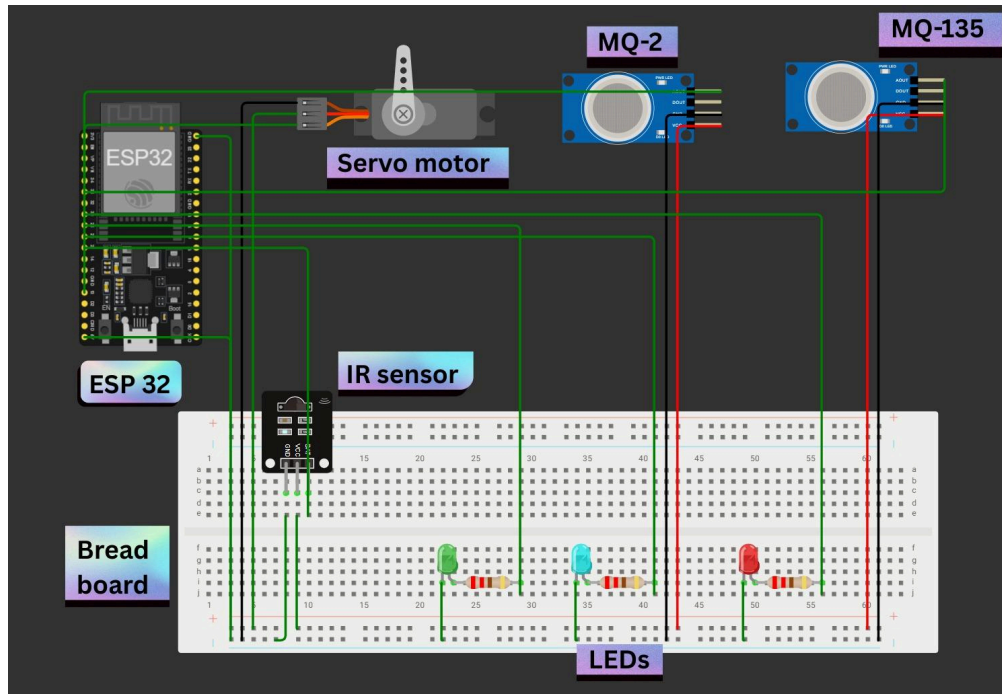
Mechanical Actuation: The Turret

Upon receiving the command packet from the Raspberry Pi, the ESP32 initiates the electromechanical response using the pan-tilt turret.

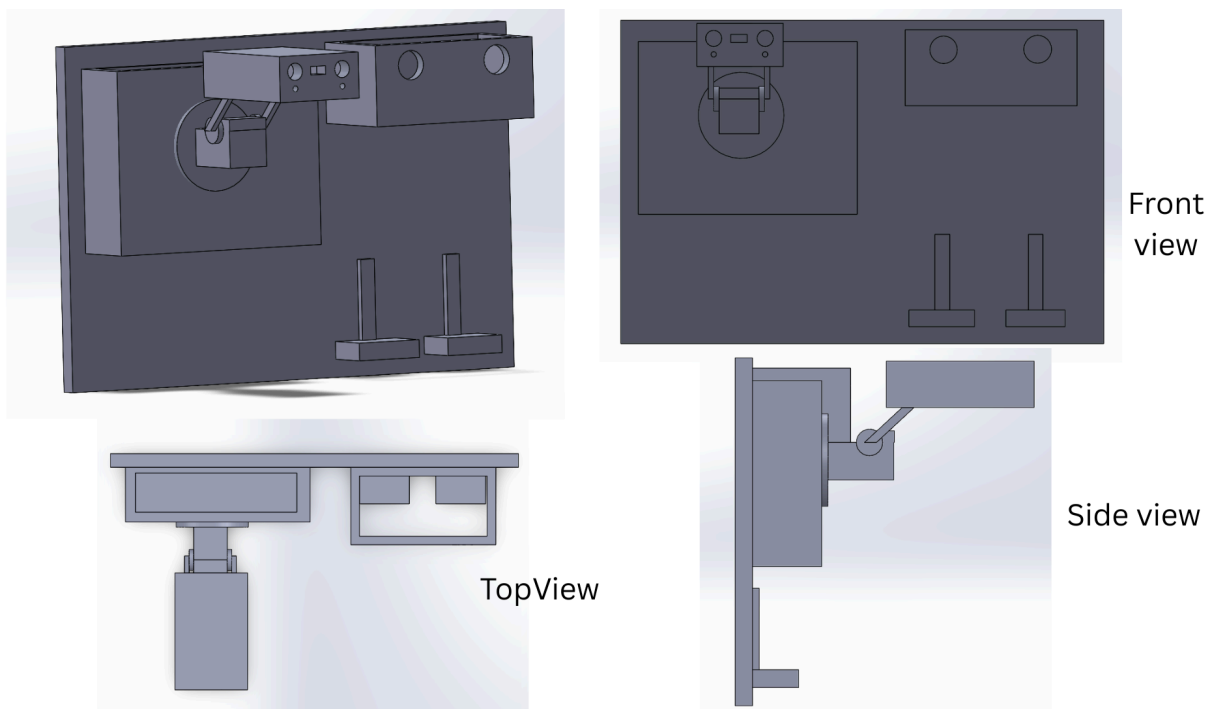
- **Dynamic Sweeping and Targeting:** The turret mechanism is constructed using one high-torque servo motors, making the turret able to rotate in one plane covering a large enough area. The ESP32 generates precise Pulse Width Modulation (PWM) signals to drive these servos. By sweeping the turret while reading real-time feedback from a localized IR sensor mounted directly on the nozzle, the ESP32 performs "peak hunting." It adjusts the PWM pulse widths until the IR sensor reads the highest intensity, ensuring the nozzle is locked precisely onto the base of the fire.
- **Agent Deployment:** Once physically locked onto the target, the ESP32 then triggers a specific channel on a multi-channel relay module. This relay acts as an electronic switch, opening the specific solenoid valve connected to the correct extinguishing agent (e.g., actuating the water pump for a Class A fire, or opening a CO-2 valve for a Class B/C fire).

Circuit Diagram

The physical integration of these components into a unified, ESP32-driven architecture requires careful attention to data protocols, signal routing, and power distribution to ensure system stability.



CAD Model



Power Distribution

Power management is centralized entirely through the ESP32. All peripheral components, including the servo motors, solenoid valves, and MQ sensors (which contain internal heating elements), draw their required current directly from the ESP32's power output pins. A common ground is established throughout the circuit to ensure all control signals and hardware components share the exact same 0 V reference point.

Data and Signal Routing:

- **Direct ADC Integration:** By utilizing the ESP32 as the sole processing unit, external Analog-to-Digital Converters are eliminated. The analog output pins of the MQ sensors and the IR sensor are routed directly to the ESP32's internal ADC pins, drastically simplifying the overall wiring footprint.
- **Logic Level Scaling:** While the ESP32 handles the AD conversion internally, it operates strictly on 3.3 V logic. To protect the ESP32's ADC pins from the higher 5 V analog signals output by the MQ sensors, simple voltage divider networks are implemented on the sensor data lines to safely scale the signals down to the 3.3 V maximum expected by the microcontroller.
- **Actuator Control:** The ESP32 outputs 3.3 V PWM (Pulse Width Modulation) signals directly to the signal wires of the pan-tilt servo motors for precise targeting. For the suppression mechanism, standard digital output pins on the ESP32 are wired to the opto-isolated input pins of the relay module. When a pin state changes, the relay electromagnetically closes the circuit, firing the appropriate extinguisher solenoid to deploy the extinguishing agent.

Cost Analysis

Prototype Costing

Serial No.	Component	Quantity	Component Cost(₹)	Total Cost (₹)
1.	ESP32 Development Board	1	450	450
2.	MQ-2 Gas Sensor	1	150	150
3.	MQ-135 Gas Sensor	1	150	150
4.	IR Flame Sensor Module	1	100	100
5.	MG996R Servo Motor	2	400	800
6.	Refillable Aluminium Cylinder	1	650	650
5.	Breadboard, Jumper Wires, Extra Batteries, LEDs, Resistors	1	400	400
	FINAL COST			2700

Commercially Scaling the Product

Scaling this automated fire suppression turret for industrial and enterprise use requires a transition to ruggedized, certified components. High-stakes environments like chemical labs and data centers demand systems that survive corrosive conditions, integrate with facility networks, and operate with zero margin for error. Consequently, the core architecture must be upgraded from basic prototyping hardware to industrial-grade infrastructure.

A critical upgrade involves the sensing and processing layers. In data centers, standard gas sensors should be replaced with Aspirating Smoke Detectors (ASD) like VESDA systems for microscopic combustion detection. Chemical labs require Multi-Spectrum IR/UV flame detectors to distinguish uncontrolled fires from controlled heat reactions. Additionally, ESP32 controllers must be replaced with Programmable Logic Controllers (PLCs) or Industrial PCs to ensure deterministic control and redundancy.

Mechanical systems also need transformation; robotic nozzles should be plumbed into high-pressure manifolds rather than relying on local canisters. This enables the deployment of massive agent reserves like Novec 1230 for electronics or high-expansion foam for chemicals. Lightweight servos must be replaced with high-torque industrial stepper motors or BLDC drives with absolute encoders for accuracy. Finally, using certified stainless steel enclosures ensures the system itself does not become an ignition source.

Market Analysis

The global fire safety equipment market is a multi-billion-dollar industry, driven by increasingly stringent safety regulations, rapid urbanization, and the modernization of commercial infrastructure. The global fire safety equipment market, valued at approximately \$60.9 billion in 2025, is projected to expand by roughly 50% over the next eight years. Parallel to this global trend, the Indian domestic market demonstrated a robust valuation of \$2.0 billion in 2025, with forecasts indicating an accelerated growth trajectory. This sustained expansion underscores a critical market transition: the continuous demand for next-generation, technologically advanced safety solutions. Given this high-growth environment and the industry's shift toward smart infrastructure, introducing an intelligent, automated system capable of both dynamic hazard classification using fusion sensors, and targeted suppression represents a highly feasible and commercially viable opportunity. However, the current market is heavily polarized. On one end of the spectrum are low-cost, purely manual solutions (handheld extinguishers). On the other end are massive, facility-wide automated systems (sprinkler networks or gas-flooding systems) that are highly expensive to install and often cause catastrophic secondary damage by deploying generic agents indiscriminately.

There is a significant, unaddressed gap in the "mid-tier" market: localized, intelligent, and highly targeted automated suppression. Our proposed autonomous turret fills this exact void. It offers the automated, 24/7 vigilance of a high-end facility system but applies the specific, localized, and agent-appropriate suppression previously only possible with a trained human responder.

The commercial application of this intelligent turret spans several distinct sectors, categorized by their inherent fire risks and the value of the assets needing protection:

- **High-Value Data Centers and Server Rooms:** These environments are highly susceptible to Class C (electrical) fires. Traditional water sprinklers are strictly prohibited, as they would destroy millions of dollars in computing equipment. An automated turret that can isolate a sparking server rack and deploy targeted CO2 without flooding the entire room is a massive commercial asset.
- **Chemical Laboratories and Manufacturing Plants:** Facilities handling volatile substances frequently face the risk of Class B (flammable liquid) and Class D (combustible metal) fires. A system capable of chemically classifying the hazard before intervening prevents explosive secondary reactions, making it ideal for hazardous material environments.
- **Commercial Kitchens and Hospitality:** Kitchens are the primary domain for Class K/F (cooking oil) fires. A localized turret mounted above cooking stations that can instantly deploy saponifying wet chemicals would drastically reduce restaurant fire damage and insurance premiums.
- **Smart Homes and Premium Residential:** With the rise of the Internet of Things (IoT) and smart home ecosystems, homeowners are increasingly investing in automated safety. A miniaturized, aesthetically pleasing version of this turret could serve as a premium upgrade to standard smoke detectors.

Future Development Roadmap

Full 2-Axis Room Coverage

The current single-axis prototype will be upgraded to a full pan-tilt dual servo mechanism, enabling 180° horizontal and 90° vertical coverage. Ceiling-mounted deployment of this configuration will eliminate blind spots and enable complete room-scale protection.

Expanded Fire Class Detection

Beyond the current Class A and Class B classification, future iterations will integrate additional sensors such as MQ-7 for carbon monoxide (Class C electrical fires) and MQ-9 for cooking oil fires (Class K), enabling the system to cover all standard fire classes and deploy the precise suppression agent for each.

Onboard Camera Integration

To eliminate laptop dependency, an ESP32-CAM module will be mounted directly on the turret for targeted real-time footage, supplemented by a wide-angle USB camera for full room coverage. This dual-camera setup ensures both close-range fire documentation and broader situational awareness without external hardware.

Computer Vision Layer

A YOLOv8 based CV model will be integrated as a secondary confirmation layer alongside the existing sensor fusion. Suppression will only trigger when both sensors and CV agree on an active flame, significantly reducing false positives from steam, sensor drift, or cooking fumes.

Wireless Monitoring

USB serial communication will be replaced with WiFi-based MQTT, enabling a real-time web dashboard for sensor data, turret status, and fire event logs. Telegram alerts will be enhanced to include fire type, timestamp, snapshot, and video clip dispatched automatically on detection.